Comprehensive Literature Review and Empirical Framework for Tabular and Deep Reinforcement Learning Architectures

The rigorous empirical comparison of reinforcement learning (RL) algorithms requires a profound grounding in both foundational theory and modern deep reinforcement learning (DRL) architecture. Transitioning from a simplistic discrete Custom GridWorld to the continuous state-space dynamics of CartPole-v1, and finally to the notoriously difficult sparse-reward challenge of MountainCar-v0, necessitates a progressive understanding of algorithmic capabilities, optimization landscapes, and fundamental limitations. This report provides an exhaustive literature review and theoretical framework spanning tabular Q-Learning, Deep Q-Networks (DQN), Double DQN, Advantage Actor-Critic (A2C), and Proximal Policy Optimization (PPO). The analysis systematically unpacks core foundational theories, algorithmic mechanics, the mathematics of reward shaping, benchmarking standards, specific reproduced pathologies, and exploration dynamics, culminating in a synthesis of hyperparameter sensitivity and the broader trajectory of the field.

SECTION 1 — Foundational RL Theory

The transition from classical optimal control to modern reinforcement learning relies on the formalization of sequential decision-making under uncertainty, modeled via Markov Decision Processes (MDPs). The foundational texts and papers in this domain establish the mathematical guarantees required to construct both tabular and deep reinforcement learning agents, dictating how value functions can be recursively estimated and optimized over time.

The theoretical bedrock of all value-based reinforcement learning rests upon the principles of dynamic programming established by Bellman in 1957.[1] In his foundational text, Bellman introduced the concept that sequential decision problems can be recursively decomposed into smaller, more manageable sub-problems. This led to the formulation of the Bellman equation, which asserts that an optimal policy possesses the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision. In the context of the reinforcement learning algorithms implemented in this research, the Bellman optimality operator provides the mathematical justification for iterative value updates. It proves that the value of a state is equal to the immediate reward plus the discounted expected value of the subsequent state, forming the recursive backbone of both tabular Q-Learning and Deep Q-Networks. Without this dynamic programming foundation, it would be impossible to mathematically guarantee that local, step-by-step updates could eventually converge to a globally optimal control policy.

While Bellman provided the exact solution methods for MDPs where the transition dynamics are fully known, it was Watkins and Dayan in 1992 who successfully adapted this framework for model-free environments.[3] Their seminal paper on Q-learning introduced the algorithm and provided its first formal convergence proof. They demonstrated that off-policy temporal difference learning converges to the optimum action-value function, Q^*, with a probability of 1. This convergence is contingent upon specific theoretical constraints: all state-action pairs must be continually updated (infinite exploration in the limit), and the learning rate must decay appropriately to satisfy stochastic approximation conditions. By proving that an agent can learn optimal policies completely independent of the behavior policy used to gather the experience, Watkins and Dayan established the framework for modern off-policy learning. This serves as the direct theoretical foundation for the tabular GridWorld implementation, verifying that despite the agent taking sub-optimal exploratory actions, the underlying Q-table will deterministically converge to the optimal control policy.

To contextualize these algorithms within the broader taxonomy of the field, Sutton and Barto's 2018 text stands as the definitive, unifying reference.[7] The textbook formalizes the complete theoretical landscape of reinforcement learning, delineating the boundaries between model-based and model-free methods, and between value-based and policy-based architectures. It comprehensively details generalized policy iteration—the interacting processes of policy evaluation and policy improvement—which governs all RL algorithms. Furthermore, the text provides the rigorous derivations for temporal-difference learning and the policy gradient theorem. The latter is of paramount importance to the advanced algorithms implemented in this dissertation, as it proves that the gradient of the performance objective with respect to the policy parameters can be computed without requiring the derivative of the state distribution. This theorem enables the direct optimization of continuous control policies seen in A2C and PPO.

The evolution of value estimation is further detailed in Sutton's 1988 publication on learning to predict by the methods of temporal differences.[8] Prior to this work, reinforcement learning heavily relied on pure Monte Carlo methods, which require waiting until the end of an episode to calculate the total return before updating the value function. Sutton introduced the $TD(\lambda)$ family of learning procedures, which assign credit by means of the difference between temporally successive predictions rather than overall episodic outcomes. This approach requires significantly less memory and peak computation while producing more accurate predictions in non-stationary environments. The introduction of the lambda return parameter mathematically bridges the gap between single-step temporal difference updates (high bias, low variance) and full Monte Carlo returns (zero bias, massive variance). This theoretical mechanism of exponentially weighting future rewards forms the direct foundation for Generalized Advantage Estimation (GAE), which is heavily utilized in modern actor-critic architectures to stabilize policy gradients.

SECTION 2 — Core Algorithm Papers

The jump from discrete, tabular spaces to high-dimensional continuous environments requires function approximation, almost exclusively via deep neural networks. The algorithms implemented for this dissertation—DQN, Double DQN, A2C, and PPO—represent distinct, critical milestones in the quest to stabilize neural network training within the highly non-stationary distributions inherent to Markov Decision Processes.

The utilization of non-linear function approximators in off-policy reinforcement learning historically resulted in catastrophic divergence, a phenomenon known as the "deadly triad." This barrier was broken by Mnih et al. in their 2015 Nature paper detailing the Deep Q-Network (DQN).[12] The authors demonstrated that deep neural networks could stably learn control policies directly from high-dimensional inputs by introducing two vital architectural components. First, they implemented an experience replay buffer, which stores past transitions and samples them uniformly during training. This breaks the temporal correlations inherent in sequential trajectory data, effectively smoothing out the training distribution and satisfying the independent and identically distributed (i.i.d.) assumptions required by stochastic gradient descent. Second, they introduced a target network—a periodically updated clone of the online network used exclusively to calculate the temporal difference target. By fixing the bootstrapping target for multiple update steps, the target network prevents the optimization landscape from continuously shifting, mitigating the risk of runaway feedback loops. These two components are strictly required for the CartPole and MountainCar DQN implementations to prevent rapid gradient divergence.

Despite its breakthrough success, the standard DQN architecture suffers from a systemic mathematical flaw. As detailed by van Hasselt, Guez, and Silver in 2016, the standard Q-learning max operator introduces a persistent overestimation bias in stochastic or noisy environments.[16] Because the target value is calculated by taking the maximum expected action-value across all available actions, and because these estimates inherently contain noise, the expectation of the maximum is strictly greater than the maximum of the true expectations (a consequence of Jensen's inequality). This causes DQN to chronically overestimate the true value of states, which can lead to suboptimal policy convergence. The authors translated the concept of tabular Double Q-Learning into the deep reinforcement learning setting to create Double DQN. This algorithm modifies the loss function by decoupling action selection from action evaluation. It evaluates the greedy policy according to the online network (selection) but calculates the value of that selected action using the target network (evaluation). This systematic elimination of positive bias prevents value functions from artificially plateauing, a critical fix implemented within the dissertation to ensure accurate value estimation in the MountainCar domain.

In the realm of policy optimization, reducing the variance of gradient estimators is essential for stable learning. Schulman et al. addressed this in 2015 with the introduction of Generalized Advantage Estimation (GAE).[20] Building directly upon Sutton's $TD(\lambda)$ framework, the paper mathematically proves that variance in policy gradient estimators can be drastically reduced at the cost of a manageable amount of bias by using an exponentially-weighted estimator of the advantage function. In actor-critic methods, the advantage function calculates how much better a specific action is compared to the baseline expectation of the state. By parameterizing this estimation with $\lambda \in $, the authors provide a mechanism to tune the bias-variance tradeoff. The paper recommends a λ value of 0.95, which provides significant variance reduction over pure Monte Carlo returns while maintaining sufficient accuracy to effectively train actor-critic architectures over long episode horizons. This mechanism is crucial for the stable training of the A2C and PPO implementations documented in the research.

The culmination of stable policy gradient optimization is represented by Proximal Policy Optimization (PPO), introduced by Schulman et al. in 2017.[24] Prior to PPO, Trust Region Policy Optimization (TRPO) was the standard for ensuring monotonic policy improvement, but it relied on computationally prohibitive second-order Hessian approximations to constrain policy updates. PPO achieves similar theoretical guarantees through a much simpler mechanism: optimizing a clipped surrogate objective. By clipping the probability ratio between the new and old policies, PPO explicitly penalizes policy updates that move the agent too far away from the behavior policy that generated the data. This clipping mechanism prevents destructively large policy updates and allows for multiple epochs of minibatch updates on the same trajectory data, vastly improving sample efficiency. Furthermore, the paper details the role of the entropy bonus in sustaining exploration across continuous and discrete control tasks, establishing PPO as the highly robust, default on-policy algorithm utilized in the dissertation's comparative analysis.

Finally, the Advantage Actor-Critic (A2C) algorithm implemented in the dissertation traces its lineage to the Asynchronous Advantage Actor-Critic (A3C) architecture published by Mnih et al. in 2016.[28] The authors proposed that parallelizing multiple actor-learners across independent environment instances serves as a natural and more memory-efficient alternative to the experience replay buffer used in DQN. By having multiple agents explore different parts of the state space simultaneously, the algorithm naturally decorrelates the stream of observed data. While the original paper focused on asynchronous gradient updates, the synchronous variant (A2C) implemented in this work waits for all parallel actors to finish their trajectory segments before calculating a unified gradient update. This paper also solidifies the standard practice of advantage normalization and multi-step bootstrapping across vectorized environments, which are essential implementation details for accelerating and stabilizing the policy gradient updates in the evaluated environments.

SECTION 3 — Reward Shaping and Sparse Rewards

The standard MountainCar-v0 environment presents a severe and classical exploration challenge. Because the agent only receives a uniform reward of -1 at each timestep regardless of its position, random exploration almost always fails to reach the top of the hill within the 200-step truncation limit. This results in a completely flat, uninformative reward signal where every generated trajectory yields an identical return of -200. To bridge this chasm of sparsity, reward shaping is frequently employed; however, altering the underlying MDP carries strict theoretical caveats and vulnerabilities.

The foundational mathematics governing reward manipulation are detailed in the 1999 paper by Ng, Harada, and Russell.[31] The authors investigate the exact conditions under which modifications to the reward function of a Markov decision process preserve the optimal policy. Their key claim and mathematical proof establish that the only form of reward shaping guaranteed to preserve the optimal policy of the original underlying MDP is potential-based reward shaping. This is strictly defined as an additive transformation $F(s, a, s') = \gamma\Phi(s') - \Phi(s)$, where Φ is an arbitrary scalar potential function defined over the state space. The paper proves that arbitrary, non-potential-based augmentations to the reward signal frequently lead to suboptimal policies, a phenomenon now commonly referred to as "reward hacking." In such failure modes, the agent learns to exploit the shaped reward signal—such as running in infinite loops to accumulate positive bonuses—rather than actually solving the primary environmental objective.

This theoretical limitation is of paramount importance to the methodology employed in the dissertation. The specific reward shaping utilized in the MountainCar experiments incorporates both height and kinetic energy, formulated as $R_{shaped} = R_{env} + c \cdot h(s) + d \cdot v(s)^2$.[35] While this heuristic strongly incentivizes the agent to build momentum and empirically accelerates learning, it is explicitly state-dependent and *not* potential-based. It violates the strict transition-difference structure required by Ng et al.'s invariance theorem. Consequently, this implementation carries an inherent limitation: if the scaling parameters c and d are improperly tuned, the agent may discover a mathematically optimal but practically useless policy. Specifically, the agent may learn to oscillate indefinitely at the bottom of the valley, harvesting continuous kinetic energy rewards without ever triggering the episode termination at the goal flag, completely overriding the original sparse task objective.[40]

The dangers of such misaligned shaping are further explored in contemporary research, such as the 2025 work by Forbes et al. on Action-Dependent Optimality-Preserving Reward Shaping (ADOPS).[43] This study highlights that "reward hacking" remains a pervasive issue when non-potential-based shaping rewards—including intrinsic motivation or manually handcrafted physical heuristics like velocity tracking—are optimized at the expense of the extrinsic environmental reward. Their work confirms the diagnosis of reward loops in environments like MountainCar, providing modern backing to the assertion that the manual insertion of velocity into the reward function is inherently vulnerable to exploitation and requires exact, delicate hyperparameter tuning to function effectively without corrupting the policy.[45]

Given these theoretical hazards, literature frequently explores alternatives to manual reward shaping. The primary modern alternative is Hindsight Experience Replay (HER), introduced by Andrychowicz et al. in 2017.[47] The key claim of HER is that the sparse reward problem can be effectively bypassed without any manual reward engineering by retroactively replaying past experiences and artificially replacing the original desired goal with the state that the agent actually achieved. By pretending that whatever state the agent reached on a failed trajectory was the intended target, the agent receives a dense, positive reward signal for that synthesized goal. This allows the agent to learn a generalized, goal-conditioned value function over the entire state space. HER provides a critical theoretical contrast to the kinetic energy rewards used in the dissertation; while the dissertation relied on modifying the MDP to guide the agent, HER demonstrates that the same problem can be solved by modifying the agent's interpretation of its own failures, preserving the purity of the original sparse reward structure.

SECTION 4 — MountainCar-v0 as a Benchmark

Within the reinforcement learning community, MountainCar serves as the quintessential diagnostic environment for analyzing exploration difficulty, temporal credit assignment, and the synthesis of momentum-building control policies.

The earliest formal description of the MountainCar problem stems from Andrew Moore's 1990 PhD thesis on efficient memory-based learning for robot control.[7] Moore's work mathematically isolates a highly specific control problem: requiring an agent to move temporarily away from its goal (by driving up the opposite, incorrect hill) in order to accrue the necessary kinetic energy and momentum to eventually reach the target. This creates a non-linear control dynamic where greedy, short-term optimization completely fails. Moore defined the continuous state space—consisting of the car's horizontal position and its velocity—as well as the discrete action space (left, neutral, right) that remains the exact mathematical standard in the modern OpenAI Gymnasium framework used in this dissertation.

Because the state space is continuous, classical Tabular Q-Learning cannot be natively applied without aggressively discretizing the environment. Logofătu et al. (2022) provide a detailed analysis of this approach in their experiments solving the MountainCar problem using state discretization.[53] By binning the continuous position and velocity values into a finite grid, the environment is coerced into a tabular format. However, the paper demonstrates that this approach exposes the algorithm to the curse of dimensionality and severe state aliasing. State aliasing occurs when distinct physical states fall into the same discrete bin, forcing the agent to treat different kinematic realities as identical. This acts as the theoretical baseline for the dissertation's tabular implementation, proving that while discretization works in principle, it places a hard, artificial ceiling on the performance ceiling compared to the smooth generalization capabilities inherent in neural network function approximators.[56]

To contextualize the empirical results achieved in the dissertation, it is necessary to establish the accepted benchmark performance levels for the evaluated algorithms on MountainCar-v0. Because the reward is -1 per step, scores closer to -100 indicate faster, highly optimized trajectories. The following table synthesizes reported solve episode counts and sample efficiency benchmarks from established repositories such as RL Baselines3 Zoo and related benchmarking literature [59]:

Algorithm	**Mean Expected Reward**	**Variance / Std Dev**	**Typical Sample Efficiency to Convergence**
PPO	-117.60 to -128.12	± 44.64	500,000 to 1,000,000 timesteps
A2C	-111.06 to	± 8.70 to	300,000 to

	-134.20	± 16.64	500,000 timesteps
DQN	-100.10 to -147.80	± 9.61 to ± 40.40	Highly variable (dependent on replay buffer)

These figures demonstrate that while DQN is capable of finding the mathematically shortest path (scores near -100) under perfect hyperparameter tuning, A2C and PPO generally converge to slightly less optimal but vastly more stable policies with significantly tighter variance bounds. These published figures provide the quantitative baseline against which the dissertation's custom implementations are rigorously compared.

SECTION 5 — Specific Pathologies Reproduced and Fixed

A central focus of the dissertation's empirical contribution is the rigorous documentation of algorithmic debugging. Implementing deep reinforcement learning involves confronting highly specific mathematical pathologies that drastically impact empirical performance. The identification, theoretical explanation, and programmatic resolution of these bugs form a critical component of the research.

Truncation Bias and the Boundary Problem

One of the most insidious bugs encountered and fixed in the A2C and PPO implementations involves the incorrect handling of episode timeouts. This pathology is formally detailed by Pardo et al. (2018) in their paper on Time Limits in Reinforcement Learning.[66] In episodic environments like MountainCar-v0, the environment forcibly truncates the episode at exactly 200 steps. In naive RL implementations, when the environment returns a done=True flag at step 200, the standard Bellman backup $Y = r_t + \gamma(1-d)V(s_{t+1})$ zeroes out the future value, treating the state exactly as if the agent had successfully hit the goal flag or died. Pardo et al. prove that this injects an artificial value cliff into the learning process; the agent learns that being alive at step 199 is inexplicably catastrophic because the value drops to zero at step 200. This corrupts the value function backward through time. The authors argue that truncations (artificial time limits) must be mathematically differentiated from true environmental terminations. The correct fix, successfully reproduced in the dissertation, involves manually checking if the termination was due to a timeout, and if so, explicitly bootstrapping the value of the final state from the critic network $V(s_{final})$ rather than zeroing it out. This restores the Markov property and prevents training instability.

Overestimation Bias in Deep Q-Networks

During the empirical evaluation, the standard DQN implementation exhibited a distinct performance plateau, averaging a score of -134 on MountainCar without further improvement. This is a direct reproduction of the mathematical pathology outlined by van Hasselt et al. in their 2016 paper.[18] As detailed in Section 2, the standard DQN calculates its target utilizing a maximization step: $Y_t^{DQN} = R_{t+1} + \gamma \max_a Q(S_{t+1}, a; \theta_t^-)$. The theoretical explanation for the plateau is that the expected value of a maximum over noisy variables is inherently greater than the maximum of the true expected values. In the highly stochastic exploration phases of MountainCar, this causes a systemic positive bias in the Q-values, warping the value landscape and causing the agent to lock onto suboptimal paths. By implementing Double DQN and decoupling the selection and evaluation metrics, this bias was mitigated, and the algorithm successfully pushed past the -134 barrier.

Entropy Regularization Sensitivity

In policy gradient methods, maintaining exploration is achieved by adding an entropy bonus to the loss function. However, the dissertation empirically found this parameter to be exceptionally sensitive: setting the entropy coefficient to 0.05 prevented the agent from consolidating a winning policy, setting it to 0.001 caused premature determinism (failing to explore the valley), while 0.01 achieved the optimal balance. This intense sensitivity is corroborated by recent literature, notably the 2025 work on Robust Policy Optimization by Cui et al..[70] The literature confirms that policy gradient methods in on-policy settings are highly vulnerable to premature convergence.[71] The entropy term $\mathcal{H}(\pi) = -\sum \pi(a|s) \log \pi(a|s)$ forces the probability distribution of actions away from deterministic extremes.[73] If the coefficient is too high, the loss function is dominated by the demand for randomness, treating the precise, sequential actions required to build momentum in MountainCar as undesirable. If too low, the network's confidence in its first random successes rapidly compounds, locking the agent into a local minimum.[75] This confirms that the empirically discovered sweet spot of 0.01 is not merely an artifact of the codebase, but a fundamental characteristic of the optimization landscape in sparse reward environments.

Replay Buffer Capacity Constraints

During the DQN ablation studies, varying the size of the experience replay buffer revealed critical constraints. A buffer of 10,000 transitions was found to be too small, causing the Q-values to diverge, while 50,000 transitions proved optimal. This aligns precisely with the findings of Zhang and Sutton in their paper "A Deeper Look at Experience Replay".[77] Their research proves that the replay buffer size is a highly sensitive hyperparameter that dictates the stationarity of the learning distribution. If the buffer is too small (e.g., 10K), the agent suffers from catastrophic forgetting; the rare, highly valuable trajectories where the agent successfully escapes the MountainCar valley are rapidly overwritten by newer, failing trajectories. Conversely, Fedus et al. (2020) demonstrate that making the buffer overly large can also degrade performance by increasing the "staleness" of the data, delaying the propagation of newly discovered, high-reward transitions into the network's gradient updates.[79] The 50K capacity thus represents the mathematically optimal tradeoff between preventing catastrophic forgetting and minimizing data staleness.

Monte Carlo vs. TD Returns Variance

A fundamental theoretical concept investigated in the dissertation is the variance of return estimators in actor-critic methods.[80] Theoretical backing dictates that calculating a full Monte Carlo (MC) return over a 200-step episode in MountainCar results in cripplingly high variance.[82] Because the environment relies on stochastic exploration to build momentum, the sheer number of compounding state transitions causes the final cumulative reward to fluctuate wildly, making gradient ascent highly unstable.[80] Conversely, a 1-step Temporal Difference (TD) return has low variance but relies almost entirely on the untrained critic network's guess, introducing immense bias.[81] The Generalized Advantage Estimation (GAE) implementation fixes this pathology by exponentially blending these n-step returns, drastically lowering the variance of the actor-critic update without relying entirely on a biased critic, providing the mathematically required stability for both PPO and A2C to converge.[82]

SECTION 6 — Exploration vs Exploitation

The fundamental tension governing all reinforcement learning agents is the exploration-exploitation dilemma. As detailed in the foundational texts by Sutton and Barto, an agent must intentionally sacrifice guaranteed short-term rewards (exploitation) to acquire novel data that enables the discovery of better long-term strategies (exploration).[7] The mechanics of how different algorithms handle this tradeoff dictates their success or failure in complex domains.

In value-based methods like Q-Learning and DQN, the standard approach is ϵ-greedy exploration. The agent takes the mathematically optimal action with probability $1 - \epsilon$, and takes a uniformly random action with probability ϵ. However, Dann et al. (2022) provide strict mathematical bounds demonstrating the limitations of this approach.[85] Their theoretical analysis proves that the sample complexity of myopic exploration policies like ϵ-greedy scales quadratically with the inverse of the exploration gap of the MDP. In environments with dense, incremental rewards like CartPole-v1, decaying ϵ from 1.0 down to 0.01 is sufficient to discover the balancing policy. However, in environments with sparse, long-horizon reward

structures, ϵ-greedy fails catastrophically.
This failure is acutely observable in the MountainCar environment. Theoretical and empirical analyses of exploration difficulty in MountainCar explain precisely why random noise is insufficient.[37] In this topology, random ϵ-greedy exploration mathematically equates to an undirected random walk in the continuous state space. Because the physical mechanics of the car require a sustained, unbroken sequence of "left" actions followed by a sustained sequence of "right" actions to build maximum kinetic energy, the probability of a random walk successfully stringing together this exact sequence without an interrupting random action is infinitesimally small.[87] Undirected noise effectively "traps" the exploration trajectory at the bottom of the gravity well. This mathematical reality necessitates the use of either intrinsic motivation, policy entropy bonuses (as in PPO), or the aforementioned kinetic energy reward shaping to artificially pull the agent's exploration away from the center of the valley.[37]

SECTION 7 — Hyperparameter Sensitivity and Reproducibility in Deep RL

The extensive hyperparameter ablation and debugging documentation generated in this dissertation highlights a systemic issue within the broader field of artificial intelligence: the reproducibility crisis in Deep RL.
This crisis was brought to the forefront by Henderson et al. in their 2018 paper, "Deep Reinforcement Learning That Matters".[89] The authors demonstrated that minor, often unreported variations in hyperparameters, random network initialization seeds, and seemingly trivial codebase implementations dictate the success or failure of Deep RL algorithms far more than the core mathematical architectures themselves. Their findings directly validate the extensive debugging methodology of this dissertation, proving that diverging results across DQN, A2C, and PPO are frequently artifacts of the hyperparameter search grid rather than fundamental proofs of algorithmic superiority.
The impact of implementation details is further localized to Proximal Policy Optimization by Engstrom et al. in their 2020 ICLR paper, "Implementation Matters in Deep Policy Gradients".[91] The authors deconstruct the empirical success of PPO and reveal that its performance is primarily driven by specific "code-level optimizations" rather than the clipped surrogate objective function alone. These hidden optimizations—which include orthogonal weight initialization for the neural networks, global reward scaling, observation normalization, and advantage normalization—fundamentally alter the numerical stability of the gradients and the scale of the advantage estimates. The paper demonstrates that without these specific tricks, PPO frequently fails to outperform vanilla, unclipped policy gradient methods. Implementing all of these optimizations was a strict requirement in the dissertation to ensure the PPO agent achieved the reported benchmark performances.
To encapsulate this ongoing challenge, Lynnerup et al. (2020) provide a comprehensive survey on reproducibility in Deep RL.[92] Their work confirms that irreproducibility is caused by a toxic combination of intrinsic algorithmic variance, environment stochasticity, and opaque

hyperparameter tuning. They advocate for rigorous statistical evaluations and standardized benchmarking frameworks to ensure fair comparisons. This survey underscores the necessity of the dissertation's ablation studies, confirming that the observed intense sensitivity in DQN buffer capacities and PPO entropy coefficients is a recognized, systemic issue across the entire discipline.[90]

SECTION 8 — Broader Context

The final phase of this literature review contextualizes the specific algorithmic implementations of the dissertation within the broader philosophical and technological landscape of modern artificial intelligence.

At a philosophical level, the entire endeavor of reinforcement learning is anchored by the "Reward Hypothesis," formally articulated by Silver, Singh, Precup, and Sutton in their 2021 paper, "Reward is enough".[95] The authors posit a provocative claim: that intelligence, and all its associated complex abilities such as knowledge representation, learning, perception, and generalization, can be understood entirely as subserving the maximization of a scalar reward signal within a given environment. This framework provides the highest-level justification for the algorithms studied in this text. It suggests that the mathematical mechanics of DQN minimizing TD error, or PPO maximizing a clipped surrogate objective, are not merely statistical optimization tricks, but potentially the core driver required to synthesize generalized artificial intelligence.

Practically, the dissertation's algorithmic spread highlights the fundamental dichotomy in modern RL research: Value-based methods versus Policy Gradient methods. The following table synthesizes the comparative theoretical strengths and weaknesses of these two primary architectures as established across the benchmarked environments [98]:

Feature	**Value-Based Methods (Q-Learning, DQN)**	**Policy Gradient Methods (A2C, PPO)**
Optimization Target	Estimates expected return of states/actions to derive a greedy policy.	Directly parameterizes and optimizes the policy network probabilities.
Sample Efficiency	Highly efficient (off-policy); data can be endlessly reused via replay buffers.	Less efficient (on-policy); requires fresh data generated by the current policy.
Action Spaces	Natively restricted to discrete, finite action spaces (requires maximization step).	Natively supports both discrete and continuous, high-dimensional action spaces.

Policy Type	Deterministic (always picks the action with highest Q-value).	Stochastic (outputs probability distributions), vital for partially observable MDPs.
Convergence	Prone to severe instability and value divergence under non-linear function approximation.	Smoother optimization landscape; stronger theoretical guarantees of monotonic improvement.

To conclude the analysis, contemporary surveys on the state of the field, such as the 2024 works by Ghasemi et al. and Korkmaz, summarize the ongoing trajectory of reinforcement learning.[102] Despite the massive deployment of DRL in complex domains ranging from medical robotics to the alignment of Large Language Models (LLMs) via RLHF, the field remains heavily constrained by the exact pathologies explored in this dissertation. Issues of sample efficiency, generalizability across different environment seeds, and extreme hyperparameter brittleness continue to plague the deployment of autonomous agents.[102] The rigorous empirical evaluation, debugging documentation, and ablation analysis of fundamental algorithms across GridWorld, CartPole, and MountainCar therefore serve not merely as an academic exercise, but as a microcosm of the exact challenges currently defining the frontier of artificial intelligence research.

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